System Validation Report

End-to-end workflows, API validation, usability and performance

Event Management System (Eventify)

1. Objective and scope

System validation checks the integrated React client, Express API and MongoDB data layer. The recorded scope includes account creation/login, roles, event lifecycle and discovery, participant registration, organizer participant views, administrator API access, security and audit behavior, usability, performance and basic data integrity.

2. Environment

- Local Ubuntu environment; Node.js v22.22.1; npm 9.2.0.
- MongoDB 7 running in Docker as a replica set (`rs0`), primary and healthy.
- Backend API on port 5000; Vite client on port 5173.
- Chrome desktop and mobile emulation (approximately 390 × 844 viewport).
- Dedicated QA accounts and event records were used; credentials are omitted.

3. Functional system validation

Test ID / feature	Validation performed	Observed result
TC-AUTH-01	Created Organizer and Participant QA accounts; authenticated and reached respective dashboards.	PASS
TC-AUTH-02	Incorrect password submitted for QA Participant account.	Rejected; remained on login page — PASS
TC-EVENT-01	Organizer created QA event.	PASS
TC-EVENT-02	Organizer edited event description, venue and capacity.	PASS
TC-EVENT-03	Organizer published event.	PASS
TC-EVENT-04	Organizer cancelled event.	PASS
TC-EVENT-05	Participant browsed event and opened event details; title/category/venue/date/capacity checked.	PASS
TC-EVENT-06	Searched by event title and filtered by Technology category and event date.	PASS
TC-REG-01	Participant registered for available capacity-one test event.	PASS
TC-REG-02	Repeated registration attempt for same participant/event.	HTTP 409; duplicate prevented — PASS
TC-REG-03	Different participant tried to join full event; event also tested with past start date.	Full event rejected; past-date event shown Closed and registration unavailable — PASS
TC-REG-04/05	Participant cancelled; history showed Cancelled and one seat became available.	PASS
TC-REG-06	Malformed event identifier submitted to registration API.	HTTP 400 — PASS
TC-ORG-01	Organizer opened participant list and saw participant name and email.	PASS
TC-ADMIN-01	Temporary Admin QA role accessed `/api/admin/users` and `/api/admin/events`.	HTTP 200 responses confirmed — PASS
TC-OPS-01	Reviewed backend application log output.	Timestamped events retrievable — PASS
TC-UX-01	Desktop workflows and mobile emulation checked for navigation, readability, controls and horizontal overflow.	PASS

4. Authorization and security validation

Check	Evidence	Result
Unauthenticated Admin API access	GET `/api/admin/users` returned HTTP 401.	PASS
Participant denied Admin users API	GET `/api/admin/users` returned HTTP 403.	PASS
Participant denied event creation	POST `/api/events` returned HTTP 403.	PASS
Organizer denied Admin events API	GET `/api/admin/events` returned HTTP 403.	PASS
Password storage	Two QA user records had bcrypt-format password hashes; no plaintext password field/value found.	PASS
Input validation	Operator-object input in email field rejected with HTTP 400 and validation message.	PASS
Error disclosure	Unknown endpoint returned HTTP 404; malformed JSON returned HTTP 400 without visible stack trace.	PASS for inspected responses

5. Audit and database validation

- Audit collection contained `event.created`, `event.updated`, `event.published` and `event.cancelled` entries for the QA event, each with `occurredAt` timestamps.
- For QA Registration Test Event: capacity = 1; stored activeRegistrations = 1; actual active registration documents = 1; withinCapacity = true; counterMatches = true.
- Registration counter consistency and capacity invariant were verified after normal register/cancel actions. A simultaneous-registration stress test was not separately executed.

6. Performance validation

Workflow	Average	p50	p95	Maximum	Threshold/result
Event listing	13.90 ms	12.67 ms	19.48 ms	29.16 ms	100/100 < 3 s — PASS
Registration	21.98 ms	20.72 ms	30.97 ms	39.28 ms	100/100 < 3 s — PASS
Text search	12.66 ms	12.26 ms	16.19 ms	18.72 ms	100/100 < 2 s — PASS
Category filter	13.06 ms	11.98 ms	18.42 ms	22.59 ms	100/100 < 2 s — PASS
Date filter	13.86 ms	12.34 ms	21.21 ms	24.40 ms	100/100 < 2 s — PASS
Combined filter	11.94 ms	11.45 ms	17.15 ms	19.46 ms	100/100 < 2 s — PASS

Each measurement script used 10 warm-up requests and 100 measured sequential requests. Temporary test events were seeded and cleaned up by the scripts. Target thresholds were at least 95% of measured requests under the relevant limit; all reported workflows achieved 100%.

7. Blocked / not separately executed

- TC-PERF-03 dashboard performance is BLOCKED: the dashboard measurement script requires a DASHBOARD_URL and no dedicated dashboard endpoint/URL was available for the test. This is not a pass.
- Concurrent-registration stress under simultaneous requests was NOT separately executed. Current evidence confirms counter consistency and capacity after sequential registration/cancellation only.
- Admin APIs were validated after temporarily assigning Admin role to a QA account; the account was restored to Organizer afterward. No user/event data was intentionally deleted.

8. Conclusion

The tested integrated workflows, role restrictions, inspected security behavior, audit records, usability checks and measured performance endpoints behaved as expected in the local QA environment. Full system-plan closure remains subject to recording a disposition for dashboard performance and completing or formally deferring concurrent-registration stress testing.